16 y is directly proportional to $\sqrt[3]{x}$

$y = 1\frac{1}{6}$ when $x = 8$

Find the value of y when $x = 64$

(Total for Question 16 is 3 marks)

17 n is an integer.

Prove algebraically that the sum of $\frac{1}{2}n(n+1)$ and $\frac{1}{2}(n+1)(n+2)$ is always a square number.

Expand the products before taking one half:

×	n	+1
n	n^2	n

×	n	+2
n	n^2	2n
+1	n	+2

1 mark
Expand both expressions

Sum = ½(n^2 + n + n^2 + 3n + 2)

= ½(2n^2 + 4n + 2) = n^2 + 2n + 1

×	n	+1
n	n^2	n
+1	n	+1

= $(n + 1)^2$

n + 1 is an integer, so its square is always a square number.

1 mark
Identify the square of an integer

(Total for Question 17 is 2 marks)

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